



Introduction to Statistical Methods

BITS Pilani

Pilani Campus

ISM TEAM



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Team ISM

Topics - Webinar



- Descriptive Statistics
 - ❖ Measures of Central Tendency
 - ❖ Measures of Variability
- Probability
 - ❖ Introduction and Basics
 - ❖ Conditional probability

Independent events

Total probability



Measures of Central Tendency

Unimodal data

Bi modal data

Tri modal data

Multimodal data

$$\frac{15+1}{2}$$

$$\frac{n+1}{2}$$

No mode

$$\text{Population } \sigma^2 = \frac{n}{n-1} \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n}$$

$$\text{Sample } s^2 = \frac{n}{n-1} \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

1) A psychologist wrote a computer program to simulate the way a person responds to a standard IQ test. To test the program, he gave the computer 15 different forms of a popular IQ test and computed its IQ from each form.

IQ Values:

134 136 137 138 138 143 144 144 145 146 146 146
 147 148 153

$n = 15$

134, 136, 137, 138, 138, 143, 144, 144, 145, 146, 146, 146, 147, 148, 153

Find the following Statistical measures:

- ✓ Mean, median, and mode
- ✓ Range, Variance and standard deviation
- iii. The interquartile range.
- v. Identify potential outliers, if any.
- vi. Construct and interpret a boxplot

Mean

$$143$$

Median = 144

Mode = 146

$$\text{Range} = L - S = 153 - 134 = 19$$

$$\text{Var} = \frac{(134 - 143)^2 + (136 - 143)^2 + \dots + (153 - 143)^2}{14}$$

$$SD = \sqrt{\text{Var}} = 5.099$$

Measures of Central Tendency

Arranging the data in ascending order:

134, 136, 137, 138, 138, 143, 144, 144, 145, 146, 146, 146, 147, 148, 153

N=15

Measure	Formula	Solution
Mean	$\mu = \frac{\sum_{i=1}^n x_i}{n}$	$\mu = 143$
Median	$p = \frac{n + 1}{2}$	Median = 144
Mode	The mode is the value or values that occur most frequently in the data set. A data set can have more than one mode, and it can also have no mode	Mode = 146

Mode, Bimodal, and Multimodal



A given set of data may have one or more than one Mode. A set of numbers with one Mode is unimodal, a set of numbers having two Modes is bimodal, a set of numbers having three Modes is trimodal, and any set of numbers having four or more than four Modes is known as multimodal.

Bimodal Mode – A set of data including two modes is identified as a bimodal model. This indicates that there are two data values that possess the highest frequencies. For example, the mode of data set B = { 8, 12, 12, 14, 15, 19, 17, 19} is 12 and 19 as both 12 and 19 are repeated twice in the given set.

No mode:

If no number in a set of numbers occurs more than once, that set has no mode: 3, 6, 9, 16, 27, 37, 48.

A unimodal mode is a set of data with only one mode.

A bimodal mode is a set of data that has two modes.

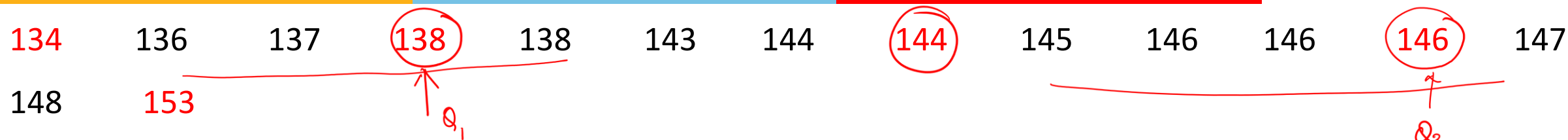
A trimodal mode is a set of data that has three modes.

Measures of Variability

Range	Range = $x_n - x_1$	Minimum = 134 Maximum = 153 Range R = 19
Variance	For a Population $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$ For a Sample $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$	Variance = 26
Standard deviation	For a Population $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ For a Sample $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$	Standard Deviation = 5.09901951

$$\begin{aligned}
 \sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\
 &= \frac{\sum (X^2 - 2\mu X + \mu^2)}{N} \\
 &= \frac{\sum X^2}{N} - \frac{2\mu \sum X}{N} + \frac{N\mu^2}{N} \\
 &= \frac{\sum X^2}{N} - 2\mu^2 + \mu^2 \\
 &= \frac{\sum X^2}{N} - \mu^2
 \end{aligned}$$

Measures of Central Tendency



- ❖ Minimum. $\rightarrow 134$
- ❖ Q1 (the first quartile, or the 25% mark). 138
- ❖ Median. $Q_2 = 144$
- ❖ Q3 (the third quartile, or the 75% mark). $\rightarrow 146$
- ❖ Maximum. $\rightarrow 153$

$$IQR = Q_3 - Q_1 = 146 - 138 = 8$$

Upper $Q_3 + 1.5IQR = 146 + (1.5)(8) = 146 + 12 = 158 \rightarrow NO$

Low $Q_1 - 1.5IQR = 138 - (1.5)8 = 138 - 12 = 126 \rightarrow NO$

$$Q_i = \text{Value of } \left(i \left(\frac{n+1}{n} \right) \right)^{th} \text{ obs., } i=1,2,3$$

Value $(3.25)^{th}$ obs

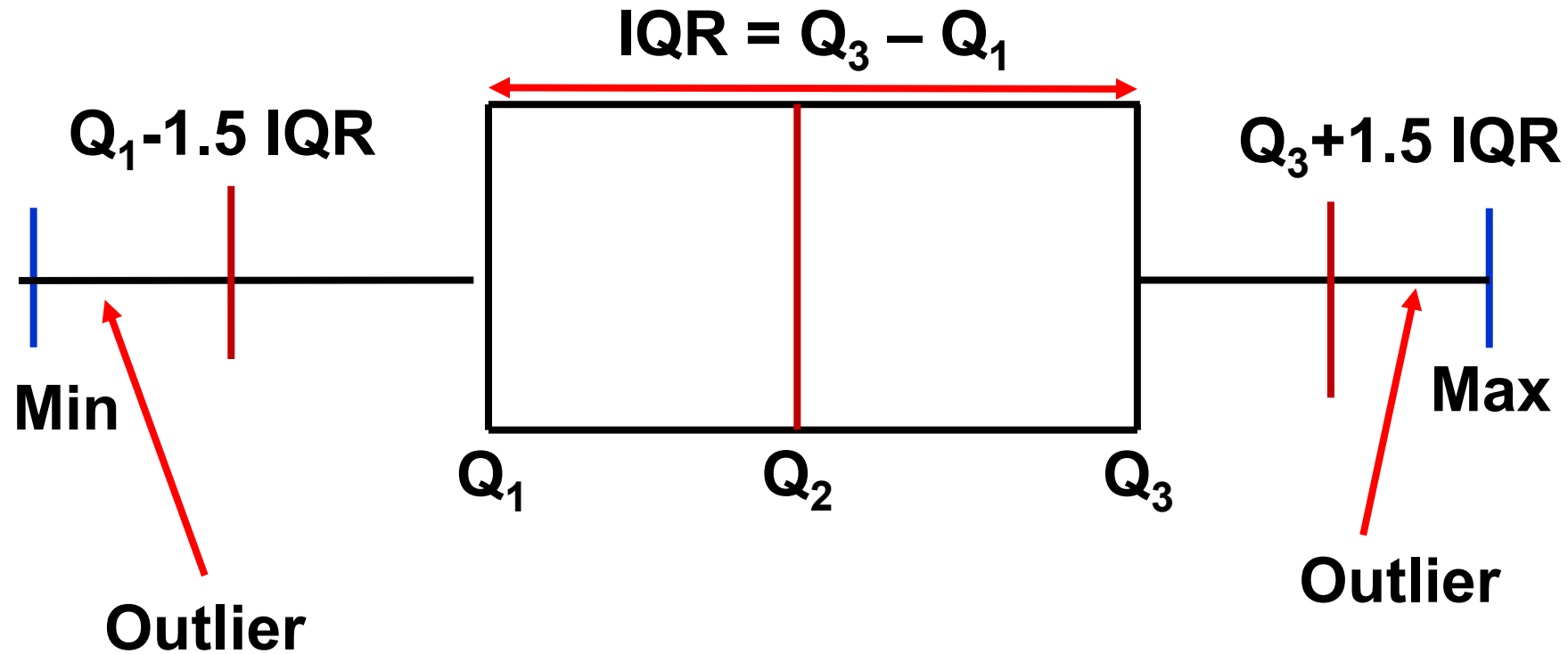
Value 3^{rd} obs $+ (0.25)(4^{th} - 3^{rd})$

Value 3^{rd} obs $+ (0.25)(\text{Value of } 4^{th} \text{ obs} - \text{Value of } 3^{rd})$

Measures of Central Tendency

Quartiles	Quartiles separate a data set into four sections. The median is the second quartile Q_2. It divides the ordered data set into higher and lower halves. The first quartile, Q_1, is the median of the lower half not including Q_2. The third quartile, Q_3, is the median of the higher half not including Q_2.	Quartiles: <table> <tr> <td>Q_1</td> <td>--></td> <td>138</td> </tr> <tr> <td>Q_2</td> <td>--></td> <td>144</td> </tr> <tr> <td>Q_3</td> <td>--></td> <td>146</td> </tr> </table>	Q_1	-->	138	Q_2	-->	144	Q_3	-->	146
Q_1	-->	138									
Q_2	-->	144									
Q_3	-->	146									
Interquartile range	$IQR = Q_3 - Q_1$	Interquartile Range $IQR = 8$									
Potential outliers, if any.	Upper Fence = $Q_3 + 1.5 \times IQR$ Lower Fence = $Q_1 - 1.5 \times IQR$	none									

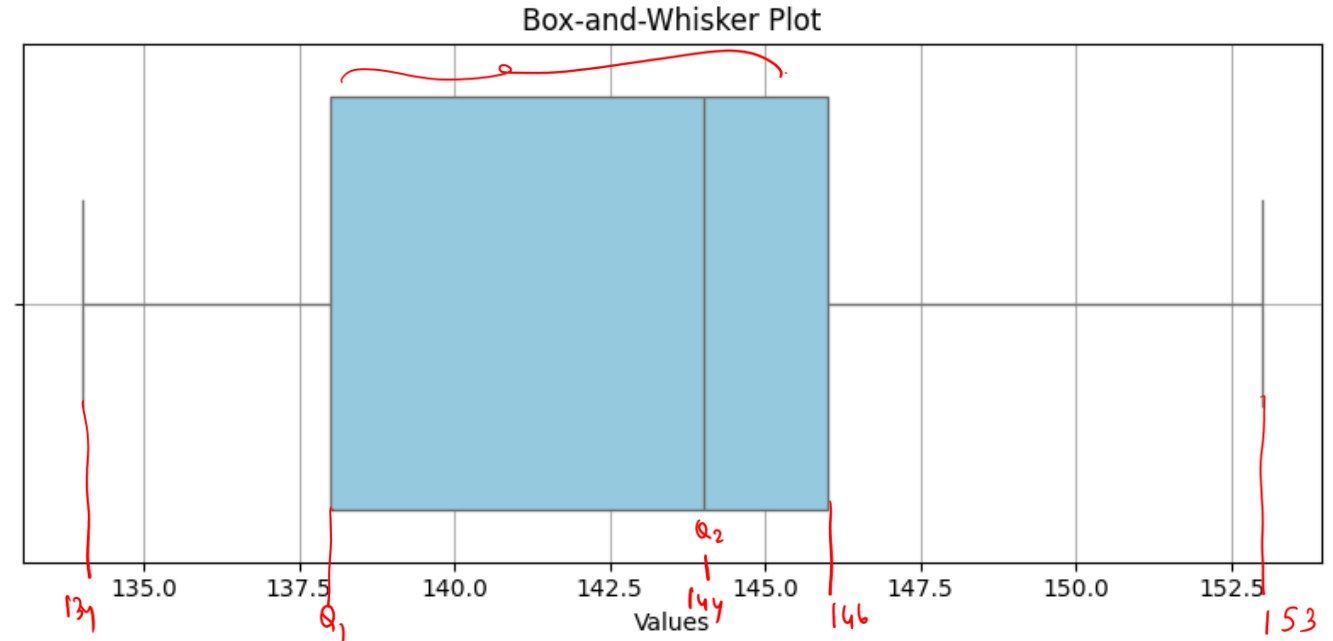
Graphical Representation: Box plot



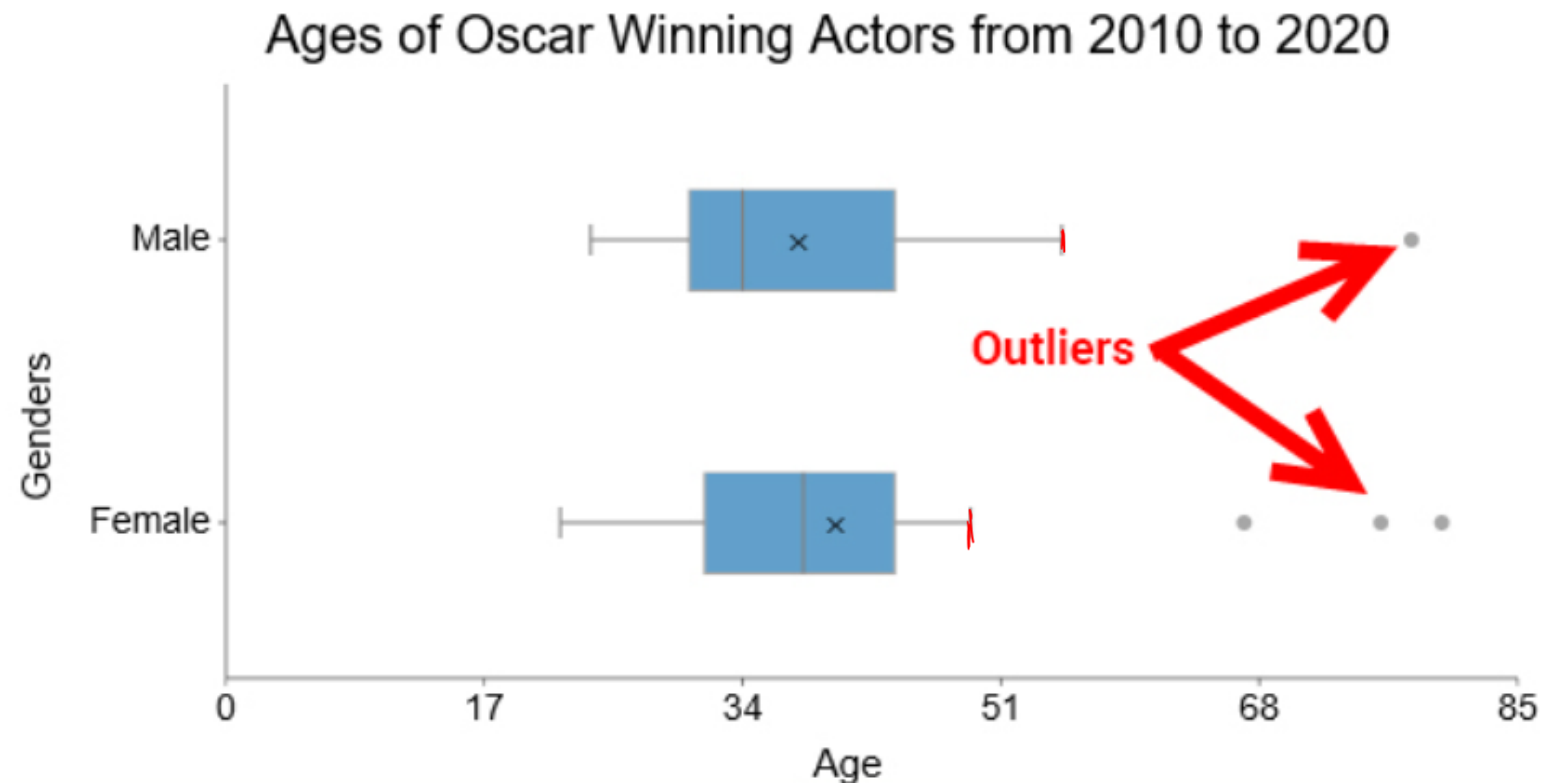
Box-plot for the given data:



Population size: 15
Median: 144
Minimum: 134
Maximum: 153
First quartile: 138
Third quartile: 146
Interquartile Range: 8
Outliers: none



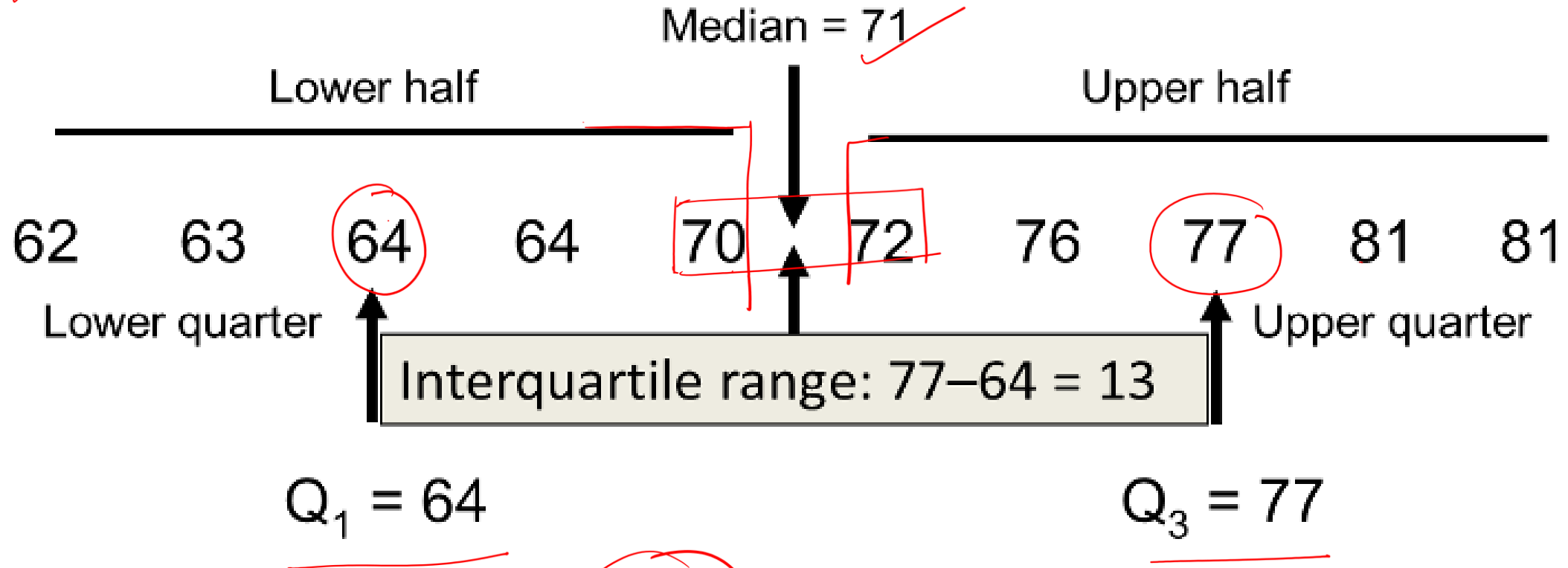
Real-time Example:



Q1 , Median (Q2) and Q3 (n is even)



$$n=10$$
$$\frac{n}{2} = \frac{10}{2} = 5$$
$$\frac{n+1}{2} = \frac{11}{2} = 5.5$$



Formula

Example:

innovate

achieve

lead

Data: 5, 7, 8, 9, 10, 12, 13, 15, 18, 30

Five-point summary:

- $Q1 = 8$ ✓
 - $Q2$ (Median) = 11 ✓
 - $Q3 = 15$ ✓ $Q_3 - Q_1$
 - $IQR = 15 - 8 = 7$ ✓
 - Lower bound = $8 - (1.5 \times 7) = -2.5$ ✓
 - Upper bound = $15 + (1.5 \times 7) = 25.5$ ✓
- ✓ Any value > 25.5 or < -2.5 is an **outlier** → here, **30** is an outlier.



2) Consider the following statistical summary of a dataset. Write at least three useful observations as a part of data pre – processing. **[Midsem Sep' 2023]**

	Nr	Cells	QValue	Fat	Protein
count	969.000000	969.000000	969.000000	969.000000	969.000000
mean	<u>7.074303</u>	<u>358.284830</u>	<u>90.016584</u>	<u>3.620279</u>	<u>3.300196</u>
std	4.759793	344.324223	4.998924	0.349956	0.136071
min	<u>1.000000</u>	<u>0.000000</u>	<u>62.650000</u>	<u>2.240000</u>	<u>2.750000</u>
25% Q_1	3.000000	157.000000	87.570000	3.410000	3.220000
50% Q_2	<u>6.000000</u>	<u>283.000000</u>	<u>90.750000</u>	<u>3.610000</u>	<u>3.310000</u>
75% Q_3	10.000000	476.000000	93.400000	3.820000	3.380000
max	<u>20.000000</u>	<u>5226.000000</u>	<u>100.000000</u>	<u>5.420000</u>	<u>3.840000</u>

$Q_3 - Q_1$
7Q2

$Sd / mean$

Contd....



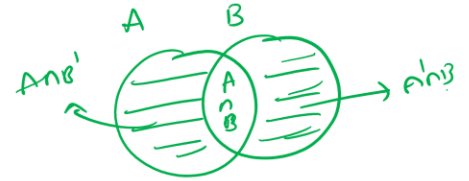
Observation 1: The variation in Protein is found very low when compared with remaining using coefficient of variation = $sd/mean$

Observation 2: The range in Protein is found very low when compared with remaining using range = $max-Min$

Observation 3: The middle range in Protein is found very low when compared with remaining using quartile range = $Q3(75\%) - Q1(25\%)$

Etc.....

Basic Probability



3) If $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/5$ then find

a). $P(A \cup B)$

b). $P(A^c \cap B)$

c). $P(A \cap B^c)$

d). $P(A^c \cap B^c)$

e). $P(A^c \cup B^c)$

f). $P((A \cup B)^c)$

$$(a) P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{5} = \frac{19}{30} = 0.6333$$

$$(b) P(A^c \cap B) = P(A' \cap B) = P(B) - P(A \cap B) = \frac{1}{3} - \frac{1}{5} = \frac{2}{15} = 0.133$$

$$(c) P(A \cap B^c) = P(A \cap B') = P(A) - P(A \cap B) = \frac{1}{2} - \frac{1}{5} = \frac{3}{10} = 0.3$$

$$(d) P(A^c \cap B^c) = P(A' \cap B') = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{19}{30} = \frac{11}{30} = 0.3667$$

$$(e) P(A^c \cup B^c) = P(A' \cup B') = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - \frac{1}{5} = \frac{4}{5} = 0.8$$

$$(f) P((A \cup B)^c) = 1 - P(A \cup B) = 1 - \frac{19}{30} = \frac{11}{30} = 0.3667$$

Solution

$$\text{a) } P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1/2 + 1/3 - 1/5 = 19/30 = 0.6333$$

$$\text{b) } P(A^c \cap B) = P(B - A) = P(B - (A \cap B)) = P(B) - P(A \cap B) = 1/3 - 1/5 = 2/15 = 0.1333$$

$$\text{c) } P(A \cap B^c) = P(A - B) = P(A - (A \cap B)) = P(A) - P(A \cap B) = 1/2 - 1/5 = 3/10 = 0.3$$

$$\text{d) } P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 19/30 = 11/30 = 0.3667$$

$$\text{e) } P(A^c \cup B^c) = P((A \cap B)^c) = 1 - P(A \cap B) = 1 - 1/5 = 4/5 = 0.8$$

$$\text{f) } P((A \mid B)^c) = 1 - P(A \mid B) = 1 - 19/30 = 11/30 = 0.3667$$



$$P(A \cap B) = P(A) \cdot P(B)$$

Basic Probability



$$P(H) = 0.01$$
$$P(B_1) = 0.02$$
$$P(B_2) = 0.02$$

$$P(\text{Saved}) = 1 - P(\text{lost})$$
$$= 1 - P(H \cap B_1 \cap B_2)$$
$$= 1 - P(H)P(B_1)P(B_2)$$

$$= 1 - (0.01)(0.02)(0.02)$$
$$= 0.999996$$

4) There is a 1% probability for a hard drive to crash. Therefore, it has two backups, each having a 2 % probability to crash, and all three components are independent of each other. The stored information is lost only in an unfortunate situation when all three devices crash. What is the probability that the information is saved?

Solution. Organize the data. Denote the events, say,

$$H = \{ \text{hard drive crashes} \},$$

$$B_1 = \{ \text{first backup crashes} \}, B_2 = \{ \text{second backup crashes} \}.$$

It is given that H , B_1 , and B_2 are independent,

$$P\{H\} = 0.01, \text{ and } P\{B_1\} = P\{B_2\} = 0.02.$$

Applying rules for the complement and for the intersection of independent events,

$$\begin{aligned} P\{ \text{saved} \} &= 1 - P\{ \text{lost} \} = 1 - P\{H \cap B_1 \cap B_2\} \\ &= 1 - P\{H\} P\{B_1\} P\{B_2\} \\ &= 1 - (0.01)(0.02)(0.02) = 0.999996. \end{aligned}$$

Basic Probability

$A_1 \rightarrow \text{Winning}$
 $A_1^c \rightarrow \text{Losing}$

$A_2 \rightarrow \text{Win}$
 $A_2^c \rightarrow \text{Lose}$

- 5) A political leader has submitted his nomination to compete in two different electoral constituencies namely A_1 and A_2 . The probability of winning in constituency A_1 and A_2 is 0.80 and 0.65 respectively. The probability of losing at least one of the constituencies is 0.35. What will be the probability that he will win in exactly one of the constituencies?

Win in A_1 & lose in A_2 $A_1 \cap A_2^c$
 Win in A_2 & lose in A_1 $A_1^c \cap A_2$

We want:

$$P(A_1 \cap A_2^c) + P(A_1^c \cap A_2)$$

$$P(A_1) - P(A_1 \cap A_2) + P(A_2) - P(A_1 \cap A_2)$$

$$P(A_1) = 0.80$$

$$P(A_2) = 0.65$$

$$P(A_1^c \cup A_2^c) = P(A_1^c \cup A_2^c) = 0.35$$

$$\Rightarrow P(\overline{A_1 \cap A_2}) = 0.35$$

$$1 - P(A_1 \cap A_2) = 0.35$$

$$P(A_1 \cap A_2) = 1 - 0.35$$

$$= 0.65$$

$$P(\text{Win exactly one}) = P(\text{Win } A_1 \text{ and lose } A_2) + P(\text{Lose } A_1 \text{ and win } A_2)$$

- $P(\text{Win } A_1 \text{ and lose } A_2) = P(A \cap B^c)$
- $P(\text{Lose } A_1 \text{ and win } A_2) = P(A^c \cap B)$

$$P(\text{He will win in exactly one of the constituency})$$

$$= P(A_1) - P(A_1 \cap A_2) + P(A_2) - P(A_1 \cap A_2)$$

$$= 0.80 - 0.65 + 0.65 - 0.65$$

$$= 0.15$$

Solution:

A political leader has submitted his nomination to compete in two different electoral constituencies namely A1 and A2. The probability of winning in constituency A1 and A2 is 0.80 and 0.65 respectively. The probability of losing at least one of the constituencies is 0.35. What will be the probability that he will win in one of the constituencies?

Assume that A, B be the events defined as follows:

A : "Winning in constituency A1"

B : "Winning in constituency A2"

Given:

$$P(A) = 0.80, P(B) = 0.65$$

$$\text{and } P(\bar{A} \cup \bar{B}) = 0.35$$

Now,

$$\therefore P(\bar{A} \cup \bar{B}) = 0.35$$

$$\therefore P(\overline{A \cap B}) = 0.35$$

$$\Rightarrow 1 - P(A \cap B) = 0.35$$

$$\Rightarrow P(A \cap B) = 0.65$$

Then, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.80 + 0.65 - 0.65$$

$$\therefore P(A \cup B) = 0.80$$

$$\begin{aligned} \text{Then, } P(\text{He will win in one of the constituencies}) &= P(A \cup B) - P(A \cap B) \\ &= 0.80 - 0.65 \end{aligned}$$

$$\therefore P(\text{He will win in one of the constituencies}) = 0.15$$

$$\begin{aligned} P(\text{He will win in constituency A1 ONLY}) &= P(A) - P(A \cap B) \\ &= 0.80 - 0.65 \\ &= 0.15 \end{aligned}$$

$$\begin{aligned} P(\text{He will win in constituency A2 ONLY}) &= P(B) - P(A \cap B) \\ &= 0.65 - 0.65 \\ &= 0 \end{aligned}$$

Independent vs. Dependent Events



Using the bag of marbles on the left, what is the probability of pulling a black marble two times in a row? $P(\text{black, black})$

When you put 1st marble back in
(Independent Events)

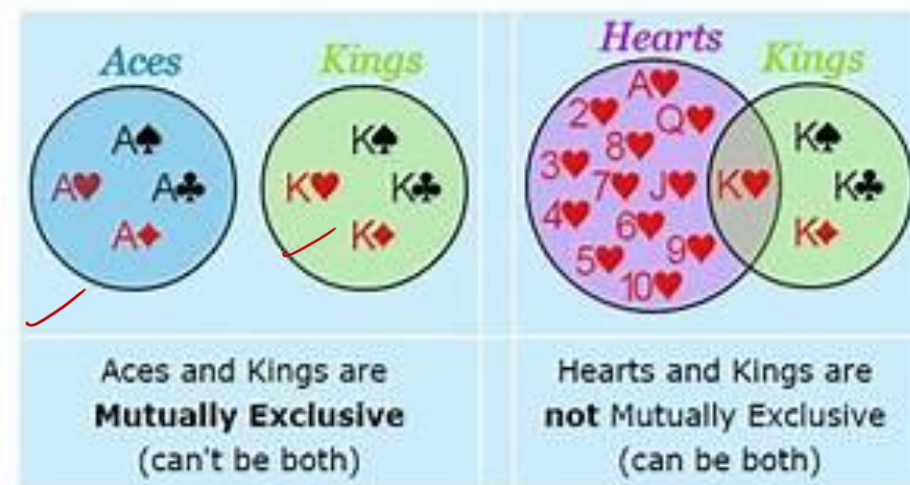
$$\frac{2}{10} * \frac{2}{10}$$

$$\frac{1}{5} * \frac{1}{5} = \frac{1}{25}$$

When you KEEP 1st marble
(Dependent Events)

$$\frac{2}{10} * \frac{1}{9}$$

$$\frac{1}{5} * \frac{1}{9}$$



Independent events

6) Comment on the statement:

“If two events are mutually exclusive, then they are independent also and vice versa”

ME $\rightarrow P(A \cap B) = 0$
I $\rightarrow P(A \cap B) = P(A)P(B)$

Two independent events cannot be mutually exclusive events - unless one or both events have a probability of zero (meaning one of the events is impossible).

Independent events

7) Let A and B be the two possible outcomes of an experiment and suppose $P(A) = 0.4$, $P(B) = p$ and $P(A \cup B) = 0.7$

(i) For what choice of 'p' are A and B mutually exclusive?

(ii) For what choice of 'p' are A and B independent?

$$\text{(i)} \quad P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.7 = 0.4 + p - 0$$

$$\Rightarrow p = 0.3$$

$$\text{(ii)} \quad P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

$$0.7 = 0.4 + p - 0.4p$$

$$0.3 = p(1 - 0.4)$$

$$0.3 = 0.6p \Rightarrow p = \frac{0.3}{0.6}$$

$$p = 0.5$$

Solution:

(i) If A and B are mutually exclusive then $P(A \cap B) = 0$

Thus $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ becomes

$$0.7 = 0.4 + P(B) - 0$$

$$\therefore P(B) = 0.7 - 0.4 = 0.3$$

(ii) If A and B are independent then $P(A \cap B) = P(A) \cdot P(B)$

Thus $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ becomes

$$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$$

$$0.7 = 0.4 + P - 0.4 \cdot P$$

$$= 0.4 + P(1 - 0.4)$$

$$= 0.4 + 0.6P$$

$$0.6P = 0.7 - 0.4 = 0.3$$

$$P = \frac{0.3}{0.6} = \frac{1}{2} = 0.5$$

$$\therefore P(B) = 0.5$$

Answer. (i) $p=0.3$, (ii) $p=0.5$



Conditional Probability

$$P(D) = 0.90 \quad P(D \cap A) = 0.75$$
$$P(A) = 0.80$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

8. Ninety percent of flights depart on time. Eighty percent of flights arrive on time. Seventy-five percent of flights depart on time and arrive on time.

- You are meeting a flight that departed on time. What is the probability that it will arrive on time?
- You have met a flight, and it arrived on time. What is the probability that it departed on time?
- Are the events, departing on time and arriving on time, independent?

Let:

- D : Flight departs on time
- A : Flight arrives on time

$$(a) \quad P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{0.75}{0.90} = \frac{75}{90} = \frac{5}{6} \approx 0.8333$$

$$\bullet \quad P(D) = 0.90$$

$$\bullet \quad P(A) = 0.80$$

$$\bullet \quad P(D \cap A) = 0.75$$

$$(b) \quad P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.75}{0.80} = \frac{75}{80} = \frac{15}{16} \approx 0.9375$$

$$(c) \quad P(A \cap B) = P(A) \cdot P(B)$$
$$P(D \cap A) = 0.75$$
$$P(D)P(A) = 0.90 \times 0.80 = 0.72$$

$\neq 0.75$
Not Independent

Solution:



- a) You are meeting a flight that departed on time. What is the probability that it will arrive on time?

$$P(A | D) = \frac{P(D \cap A)}{P(D)}$$
$$P(A | D) = \frac{0.75}{0.90} = \frac{75}{90} = \frac{5}{6} \approx 0.833$$

- b) You have met a flight, and it arrived on time. What is the probability that it departed on time?

$$P(D | A) = \frac{P(D \cap A)}{P(A)}$$
$$P(D | A) = \frac{0.75}{0.80} = \frac{75}{80} = \frac{15}{16} = 0.9375$$

c) Are the events, departing on time and arriving on time, independent?

Two events A and B are independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Check:

$$P(D) \cdot P(A) = (0.90)(0.80) = 0.72$$

But:

$$P(D \cap A) = 0.75 \neq 0.72$$

So, the events are NOT independent.

Conditional Probability

$$P(F) = 0.30$$

$$P(A) = 0.45$$

$$P(F \cap A) = 0.05$$

9) In an online shopping survey, 30% of persons made shopping in Flipkart, 45% of persons made shopping in Amazon and 5% made purchases in both. If a person is selected at random, find

- the probability that he makes shopping in at least one of two companies $\rightarrow U$
- the probability that he makes shopping in Amazon given that he already made shopping in Flipkart.
- the probability that the person will not make shopping in Flipkart given that he already made purchase in Amazon.

$$i) P(F \cup A) = P(F) + P(A) - P(F \cap A) = 0.3 + 0.45 - 0.05 = 0.7$$

$$ii) P(A / F) = \frac{P(A \cap F)}{P(F)} = \frac{0.05}{0.30} = 0.167$$

$$iii) P(F^c / A) = \frac{P(F^c \cap A)}{P(A)} = \frac{P(A) - P(F \cap A)}{P(A)} = \frac{0.45 - 0.05}{0.45} = \frac{0.40}{0.45} = 0.888$$

Conditional Probability

Solution: Given $P(F) = 30\% = 0.30$

$$P(A) = 45\% = 0.45$$

$$P(F \cap A) = 5\% = 0.05$$

$$\text{i) } P(F \cup A) = P(F) + P(A) - P(F \cap A)$$

$$= 0.30 + 0.45 - 0.05 = 0.7$$

$$\begin{aligned} \text{ii) } P(A | F) &= \frac{P(A \cap F)}{P(F)} \\ &= \frac{0.05}{0.30} = 0.167 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(F' | A) &= \frac{P(F' \cap A)}{P(A)} \\ P(F' \cap A) &= P(A) - P(A \cap F) \\ &= 0.45 - 0.05 \\ &= 0.40 \end{aligned}$$

$$P(F' | A) = \frac{0.40}{0.45} = 0.88$$



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B) P(B)$$

$F \rightarrow \text{Faulty}$

Total Probability

10) A businessman goes to hotels X, Y, Z -20%, 50%, 30% of the time, respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbing. Determine the probability that the businessman goes to hotel with faulty plumbing.

Given:

Hotel Visit Probabilities:

- $P(X) = 0.20$
- $P(Y) = 0.50$
- $P(Z) = 0.30$

Probability of Faulty Plumbing (given hotel):

- $P(F|X) = 0.05$
- $P(F|Y) = 0.04$
- $P(F|Z) = 0.08$

$$P(X) = 0.20$$

$$P(Y) = 0.50$$

$$P(Z) = 0.30$$

$$P(F|X) = 0.05$$

$$P(F|Y) = 0.04$$

$$P(F|Z) = 0.08$$

$$\begin{aligned} P(F) &= P(F \cap X) + P(F \cap Y) + P(F \cap Z) \\ &= P(F|X) \cdot P(X) + P(F|Y) P(Y) + P(F|Z) P(Z) \\ &= (0.05)(0.20) + (0.04)(0.50) + (0.08)(0.30) \\ &= \underline{\underline{0.054}} \end{aligned}$$

Find the **total probability** that the businessman goes to a hotel with faulty plumbing:

$$P(F) = P(X) \cdot P(F|X) + P(Y) \cdot P(F|Y) + P(Z) \cdot P(F|Z)$$

$$P(F) = (0.20 \times 0.05) + (0.50 \times 0.04) + (0.30 \times 0.08)$$

$$= 0.010 + 0.020 + 0.024 = \boxed{0.054}$$

Total Probability

11) A company receives 50% of customer complaints via phone, 30% via email, and 20% via chat.

Resolution rates:

- Phone: 90%
- Email: 80%
- Chat: 70%

What is the probability that a randomly selected complaint is resolved?

Given:

- $P(\text{Phone}) = 0.50$ (50% of complaints come via phone),
- $P(\text{Email}) = 0.30$ (30% of complaints come via email),
- $P(\text{Chat}) = 0.20$ (20% of complaints come via chat),
- $P(\text{Resolved} | \text{Phone}) = 0.90$ (90% of phone complaints are resolved),
- $P(\text{Resolved} | \text{Email}) = 0.80$ (80% of email complaints are resolved),
- $P(\text{Resolved} | \text{Chat}) = 0.70$ (70% of chat complaints are resolved).

$R \rightarrow \text{Resolved}$

$$P(\text{Phone}) = 0.50$$

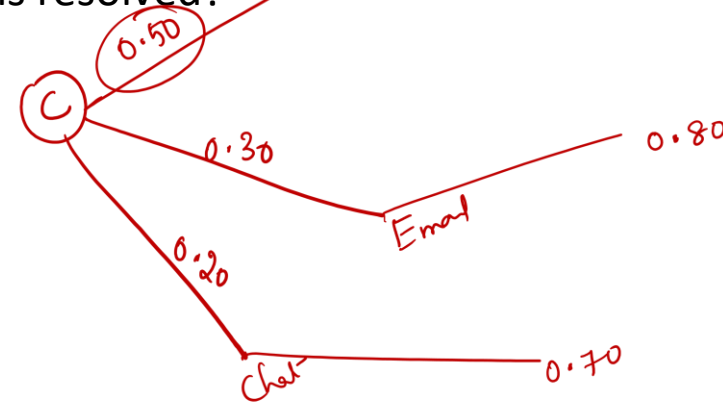
$$P(\text{Email}) = 0.30$$

$$P(\text{Chat}) = 0.20$$

$$P(R | \text{Phone}) = 0.90$$

$$P(R | \text{Email}) = 0.80$$

$$P(R | \text{Chat}) = 0.70$$



$$P(R) = P(R \cap \text{Phone}) + P(R \cap \text{Email}) + P(R \cap \text{Chat})$$

$$= (0.50)(0.90) + (0.30)(0.80) + (0.20)(0.70)$$

$$= 0.83$$

To calculate $P(\text{Resolved})$, we'll use the **Law of Total Probability**. The formula is:

$$P(\text{Resolved}) = P(\text{Phone}) \cdot P(\text{Resolved} \mid \text{Phone}) + P(\text{Email}) \cdot P(\text{Resolved} \mid \text{Email}) + P(\text{Chat}) \cdot P(\text{Resolved} \mid \text{Chat})$$

Solution:

$$P(\text{Resolved}) = 0.50 \cdot 0.90 + 0.30 \cdot 0.80 + 0.20 \cdot 0.70 = 0.45 + 0.24 + 0.14 = 0.83$$

Conclusion:

So, there is an **83%** chance that a randomly selected customer complaint will be resolved, based on the given resolution rates for each channel.

Problem:

$$P(G) = \frac{7}{12}, \quad P(B) = \frac{5}{12}$$

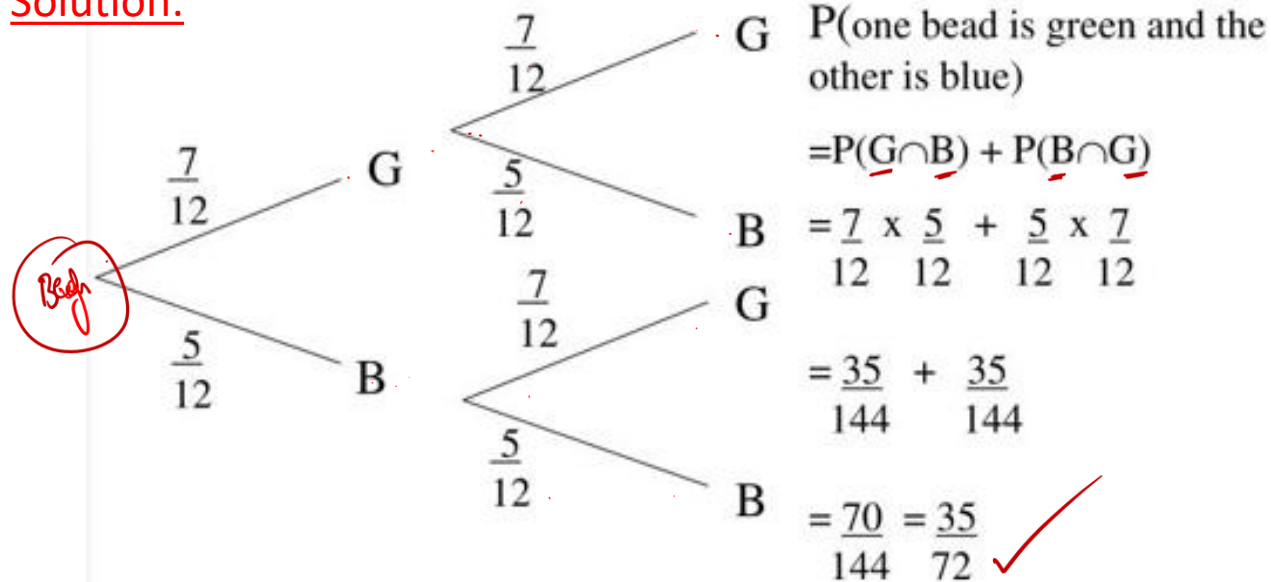
with replacement
Independent

12.

A bag contains 7 green beads and 5 blue beads. A bead is taken from the bag at random, the colour is recorded and the bead is replaced. A second bead is then taken from the bag and its colour is recorded.

- Find the probability that one bead is green and the other is blue
- Show that the event “the first bead is green” and “the second bead is green” are independent

Solution:



$$P(G \cap B) = P(G)P(B)$$
$$P(B \cap G) = P(B)P(G)$$

$$P(G \cap G) = P(G)P(G)$$

Problem:

$$P(A \cap B \cap C) = 0$$

$$\begin{aligned} P(B \cup C) &= P(B) + P(C) - P(B \cap C) \\ &= P(B) + P(C) \\ &= \frac{5}{10} + \frac{4}{10} = \frac{9}{10} = 0.9 \end{aligned}$$

13. Suppose A, B and C are three mutually exclusive events in a sample space. Given that $S = A \cup B \cup C$, $P(A) = (1/5)P(B)$, and $P(C) = 4P(A)$, find $P(B \cup C)$.

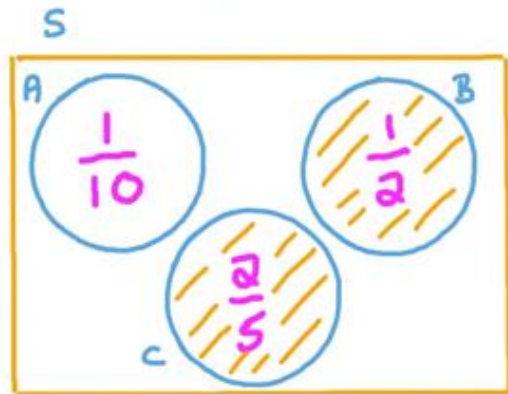
Solution:

$$\begin{aligned} P(B) &= 5P(A) & P(C) &= 4P(A) \Rightarrow \frac{P(C)}{P(A)} = \frac{4}{1} \\ P(B) &= \frac{5}{10} \end{aligned}$$

$$\begin{aligned} P(S) &= P(A) + P(B) + P(C) \\ \Rightarrow 1 &= P(A) + 5P(A) + 4P(A) \\ \Rightarrow 10P(A) &= 1 \Rightarrow P(A) = \frac{1}{10} \end{aligned}$$

$$5P(A) = P(B)$$

Two or more events are mutually exclusive if they cannot happen at the same time. $P(X \cup Y) = P(X) + P(Y)$



$$\begin{aligned} P(A \cup B \cup C) &= 1 \\ P(A) + P(B) + P(C) &= 1 \\ P(A) + 5P(A) + 4P(A) &= 1 \\ 10P(A) &= 1 \\ P(A) &= \frac{1}{10} \end{aligned}$$

$$\begin{aligned} P(B \cup C) &= P(B) + P(C) \\ &= \frac{5}{10} + \frac{4}{10} \\ &= \frac{9}{10} \end{aligned}$$

Below the equations, arrows point to the final values: $P(B) = \frac{5}{10} = \frac{1}{2}$ and $P(C) = \frac{4}{10} = \frac{2}{5}$.

$$\begin{aligned} S &= A \cup B \cup C \\ P(S) &= P(A \cup B \cup C) \\ 1 &= P(A) + P(B) + P(C) \end{aligned}$$

Practice Problems

1. If $P(A) = 1/3$, $P(B) = 1/2$, $P(A/B) = 1/6$ find i). $P(B/A)$ ii). $P(B/A')$ iii). $P(A \cup B / A)$ iv). $P(B/A)$.

2. Three machines A, B, C produce 50%, 30%, and 20% of the items in a factory. The percentage of defective outputs of these machines are 3, 4 and 5 respectively.
 - (i) If an item is selected at random, what is the probability that it is defective? (Ans: 0.037)
 - (ii) If a selected item is defective, what is the probability that it is from machine A? (Ans: 0.4054)



Practice Problems

3. A weather app predicts rain with 90% accuracy when it actually rains. It wrongly predicts rain 10% of the time when it doesn't rain. Suppose it rains only 10% of days in a year. You check the app and it predicts rain. What is the probability it will actually rain? (Ans: 0.5)

4. Consider the following data related to the employees, who are on travel. 40% check work email, 20% use cell phone to stay connected to work, 25% bring laptop with them, 23% check both work email and use cell phone to stay connected, and 50% neither check work email nor use a cell phone to stay connected nor bring a laptop. In addition, 88 out of every 100 who bring a laptop also check work email, and 70 out of every 100 who use a cell phone to stay connected also bring a laptop.

- i) What is the probability that a randomly selected traveller who checks work email also uses a cell phone to stay connected?
- ii) What is the probability that someone who brings a laptop on vacation also uses a cell phone to stay connected?
- iii) If the randomly selected traveller checked work email and brought a laptop, what is the probability that he/she uses a cell phone to stay connected?

TEXT BOOKS

T1 : Statistics for Data Scientists, An introduction to Probability,
Statistics and Data Analysis, Maurits Kaptein et al, Springer 2022

T2 : Probability and Statistics for Engineering and Sciences,
8th Edition, Jay L Devore, Cengage Learning

T3 : Introduction to Time Series and Forecasting, Second Edition,
Peter J Brockwell, Richard A Davis, Springer.

Questions please...





THANK YOU!